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Cryptography

- Practice of protecting your data or information by transforming it into secure format.

- Uses / Application :

  To prevent data leakage.
  To maintain confidentiality.
  Prevent unauthorized access.

1] Symmetric : Both keys are same.

2] Asymmetric : i) private key ii) public key.

- Cryptology is study of cryptography & crypt analysis.

- Crypt Analysis is Breakdown the Encryption.

* Classical Encryption Techniques :

1] Ceasar Cipher :

It is a substitution cipher where each letter in the plain text is shifted by a fixed no. of positions in the alphabets.

eg: AB → CD
HI → JK.

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2] Substitution Cipher:
Each letter or group of letter in the plain text is replaced by another letter or symbol majorly uses keyword to shift the alphabet.

3] Transposition Cipher:
It involves rearranging the character in the plain text rather than substituting them

eg: HELLO → OLLEH.

* Playfair cipher :

- To create a 5x5 matrix of letters is called Grid of letter.
- The matrix is made by inserting the values of key & remaining alphabets into the matrix row wise from left to right.
- When letter I & J are compiled together convert the plain text into pairs of alphabet.

i) pairs cannot be made with same words.
ii) break the letter into pairs of alphabet.

- If the letter is standing alone in the process of pairing then add z in the letter.

1] If both alphabets in same row replace them & with alphabets to their immediate right.

2] If both alphabets are in same column then replace

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them with alphabets immediate below them.

3] If not in the same row & column replace them with alphabets in same row respectively but add other pair of corners.

eg:

	1	2	3	4	5
1	D	I	V	Y	A
2	B	C	E	F	G
3	H	K	L	M	N
4	O	P	Q	R	S
5	T	U	W	X	Z

KN = LH
JY = VA
FM = MR
CV = KI
CR = FP
HS = NO
DW = VT.

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* Hill cipher :

Encrypts a group of letter called polygraph like in playfair cipher we were encrypting pair of letters called as diagraphs.

$C = KP \bmod 26$

C = cipher text , K = key , P = plain text

eg: $K = DEAD = \begin{bmatrix} D & E \\ A & D \end{bmatrix}_{2\times2} = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix}$

$P = ATTACK = \begin{bmatrix} A \\ T \end{bmatrix}_{2\times1} \begin{bmatrix} T \\ A \end{bmatrix} \begin{bmatrix} C \\ K \end{bmatrix}$

1] $C = KP \bmod 26$

$C = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} \cdot \bmod 26$

$\begin{bmatrix} 3\times0 + 4\times19 \\ 0\times0 + 3\times19 \end{bmatrix}$

$= \begin{bmatrix} 76 \\ 57 \end{bmatrix} = \begin{bmatrix} 24 \\ 5 \end{bmatrix}$

$\begin{pmatrix} 76 \\ 57 \end{pmatrix}$

$= \begin{bmatrix} Y \\ F \end{bmatrix} = \underline{\underline{YF}}$

mod 26 = divide by 26.

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2] $C = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 19 \\ 0 \end{bmatrix} \cdot \text{mod } 26$

$= \begin{bmatrix} 57 \\ 0 \end{bmatrix} \cdot \text{mod } 26$

$= \begin{bmatrix} 5 \\ 0 \end{bmatrix} = \begin{bmatrix} F \\ A \end{bmatrix} = \underline{\underline{FA}}$

3] $C = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 10 \end{bmatrix} \cdot \text{mod } 26$

$= \begin{bmatrix} 46 \\ 30 \end{bmatrix} \cdot \text{mod } 26$

$= \begin{bmatrix} 20 \\ 04 \end{bmatrix} = UE$

= YFFAUE

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A=0
B=1
C=2
D=3
E=4
F=5
G=6
H=7
I=8
J=9
K=10
L=11
M=12
N=13
O=14
P=15
Q=16
R=17
S=18
T=19
U=20
V=21
W=22
X=23
Y=24
Z=25

* Transposition Cipher :

[Note : one time pad is not a transposition cipher it is substitution cipher].

Varnam :

Length of the key used for encryption should be equal to length of plain text.

plain text = L O V E
Key = L I F E [encryption]

i.e 11 14 21 4
\+ 11 8 5 4

22 22 26 8
W W A I

∴ WWAI = cipher text.

W W A I
\- L I F E
22 22 26 8 decryption.
\- 11 8 5 4
11 14 21 4
L O V E

For Encryption = Add
For Decryption = Substract.

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Plain text = SECURITY
Key = COMPUTER

S E C U R I T Y
+ C O M P U T E R

18 4 2 20 17 8 19 24
+ 2 14 12 15 20 19 4 17
20 18 14 35 37 27 23 41

U S O J L B X P = C.T

Now Decryption :

U S O J L B X P
- C O M P U T E R

20 18 14 35 37 27 23 41
- 2 14 12 15 20 19 4 17
18 4 2 20 17 8 19 24

S E C U R I T Y.

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- Types :

1] Rail Fence Transposition
2] Columnar Transposition
3] Row Transposition.

Imp Q] Diff betn substitution & Transposition.

1] Rail Fence -

i) Key = INFORMATION

I F R A I N

N O M T O

Cipher text = IFRAINNOMTO

ii) Key = AUTHENCITY AUTHENTICITY

A T E T C T

U H N I I Y

Cipher text = ATETCTUHNIIY

2] Columnar : matrix is used.

1] Plain text = LET US MEET

L	E	T
U	S	M
E	E	T

3×3 LUEESETMT

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- Note: Read cipher text from top to down.

ii) INFORMATION TECHNOLOGY

I	N	F	O
R	M	A	T
I	O	N	T
E	C	H	N
O	L	O	G
Y			

4x6

IRIEOYNMOCLFANHOOTTNG = cipher text

3] Row Transposition:

In this type, a key will be given & according to it the cipher text will be written. The key will be numbers / sequence of row in matrix / table.

eg: Key = 312

1	2	3
L	E	T
U	S	M
E	E	T

∴ Cipher text = TMTLUEESE

The cipher text will be found according to the key seq. given.

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- Plain text = SUN IS BRIGHT
  Key = 53412

1	2	3	4	5
S	U	N	I	S
B	R	I	G	H
T				

∴ Cipher text = SHNIIGSBTUR.

* Hashing :
  It is a process to convert input data into a fixed string of characters called as hash value or hash code.

* Steganography :
  It is a practice of concealing the information in a way that prevents it's detection unlike encryption that scrambles the data to make it unreadable without a key.
  It can be done in images, text, audio, video.

1] Image steganography : Changes least significant bit.
   It works on pixels.

2] Audio steganography :
   Changing frequency or altering wave patterns.

3] Text steganography :
   Hide messages using spaces, emojis, changing font size, style variations.

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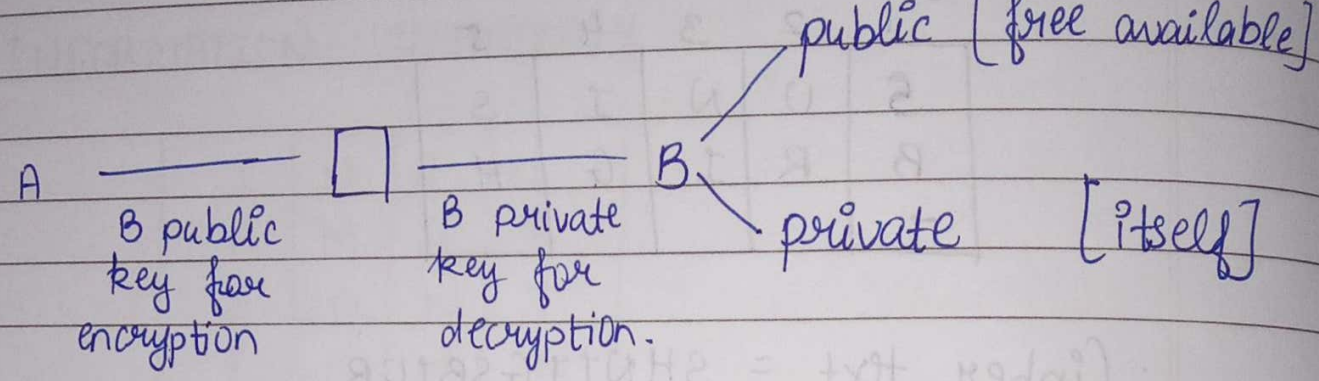
- Asymmetric key:



- Algorithms:

- RSA
- ECC - elyptic curve cryptography
- DSA - digital signature algorithm.

- Symmetric key: Both keys are same
- Algorithm:

- DES
- 3DES
- AES

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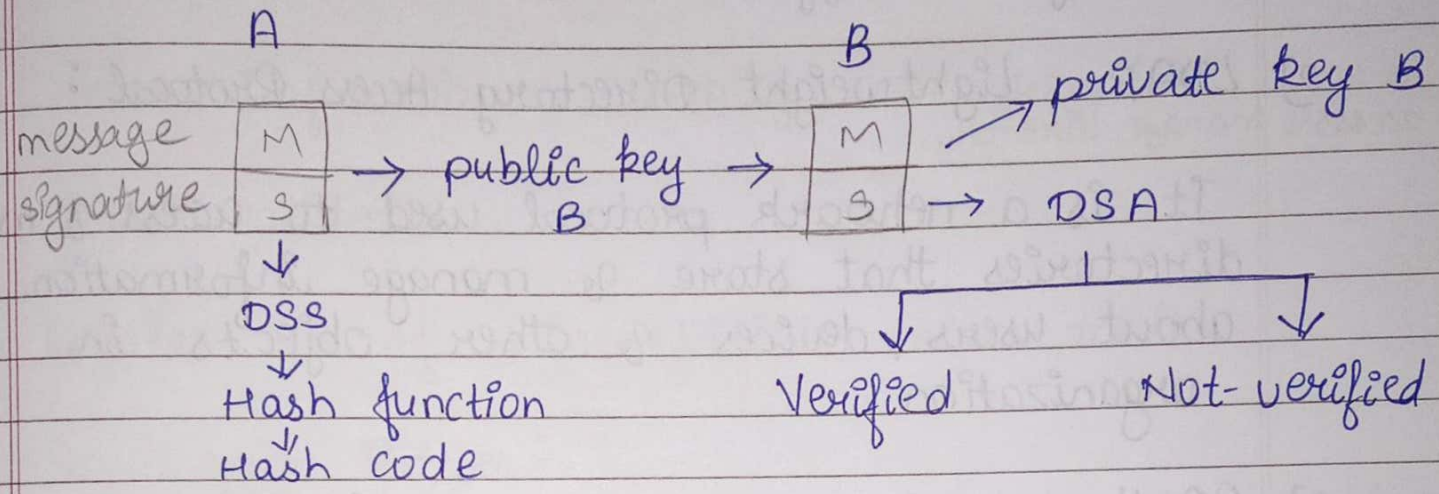
* Digital Signature :

Digital signature is a verification method. It does not provide confidential communication.



* Digital Signature Standard :

DSS is a federal information processing standard which defines algorithm that are used to generate digital signatures with the help of SHA - [secure hash algorithm] for the authentication of electronic documents.

DSS is a standard that marks digital signature robust.

Q] Diff betn symmetric & asymmetric.
Q] DSS Working ?
Q] What is steganography.

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* Authentication Protocols :

1] Kerberos : identify users & devices on n/w.

Authentication protocol that is used to securely identify users & devices on a network.

2] LDAP - Lightweight Directory Access Protocol :
access & manage directory.

It is a network protocol used to access & manage directories that store & manage information about users, devices & other objects in organization.

3] OAuth.2 - Open Authorization .2 : OTP
grant third party applicatn access to resources without sharing pswd for specific timing.

It is an open standard that enables users to grant third party application access to their resources without sharing their password for a specific timing.

4] SAML - Security Assertion Markup Language :
exchange authentication & authorizatn data betn organizations

It is a networking or standard protocol that is used to securely exchange authentication and authorization of data between organizations.

5] RADIUS - Remote Authentication Dial In user Service :

It is used to manage & authenticate user who connect to a network.